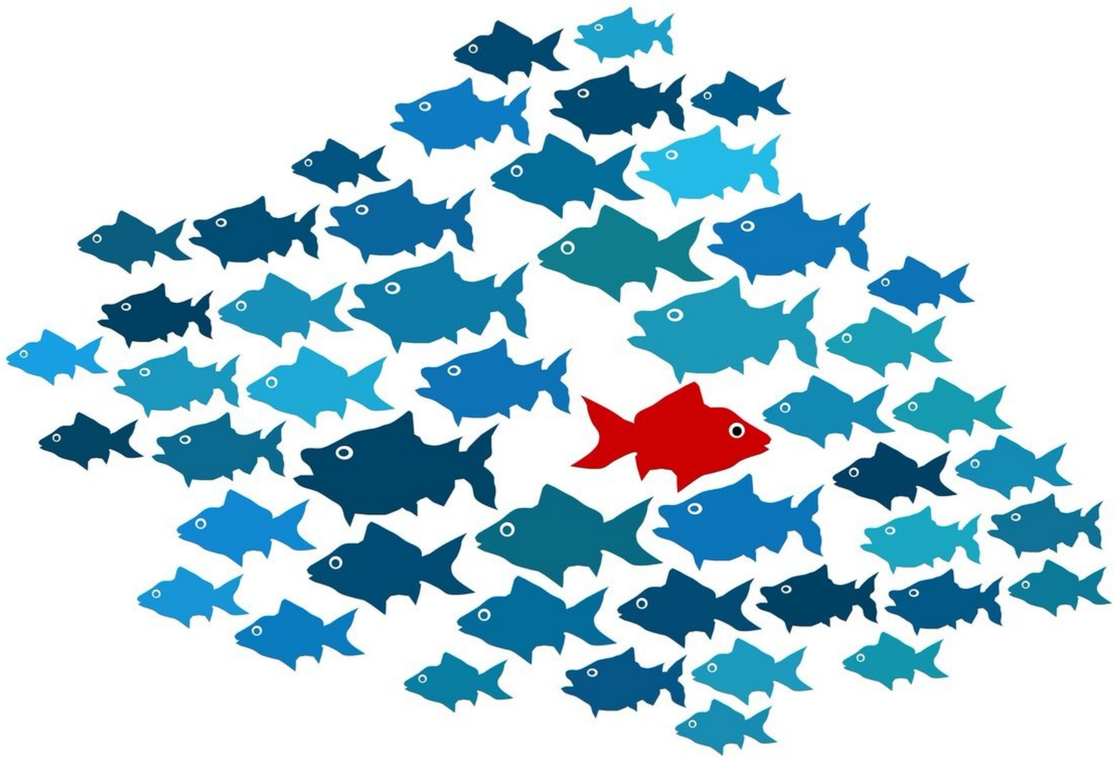
Detección de Anomalías

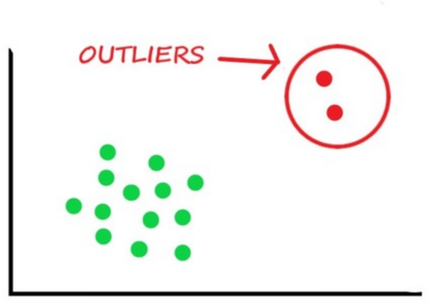
Valero Laparra
Departament d'Enginyeria Electrònica,
Escola Tècnica Superior d'Enginyeria
Valero.laparra@uv.es

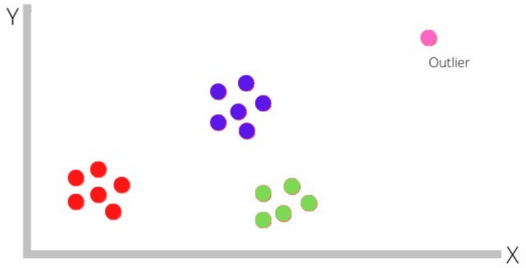
INTRODUCCIÓN

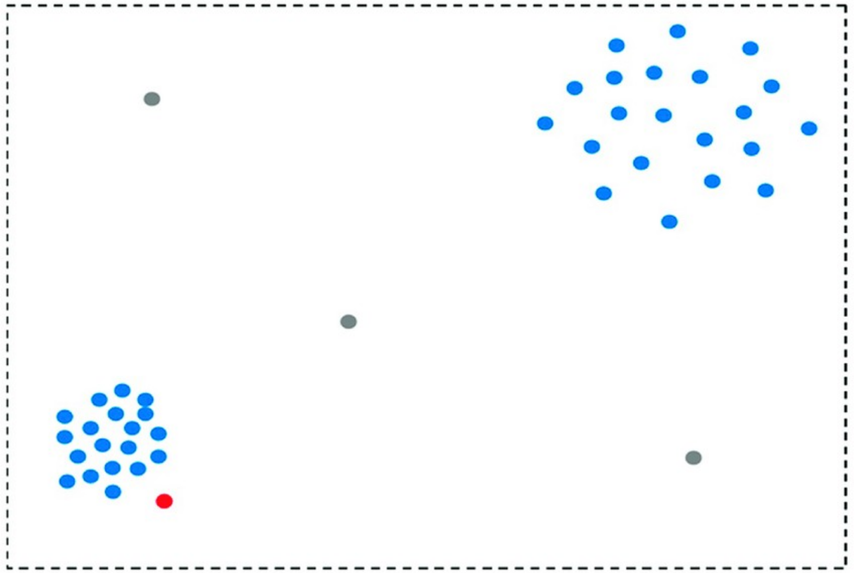


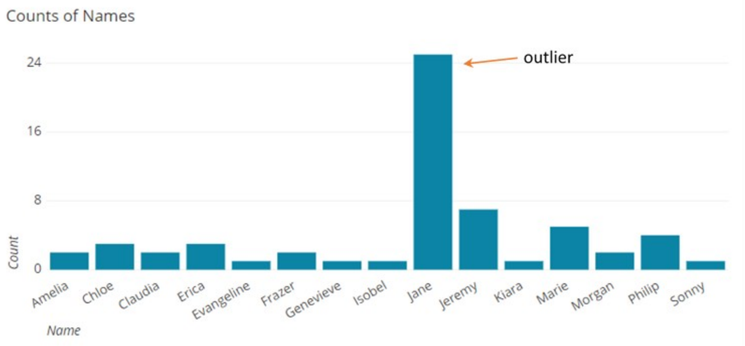
INTRODUCCIÓN

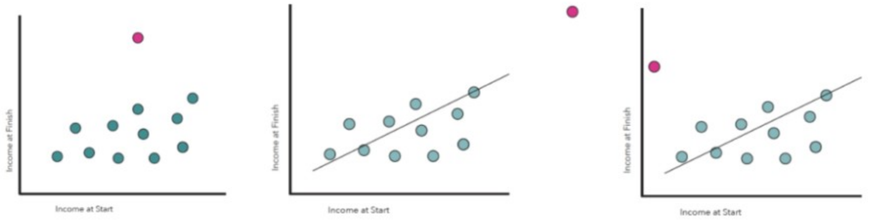
Anomalia/Outlier/Ruido



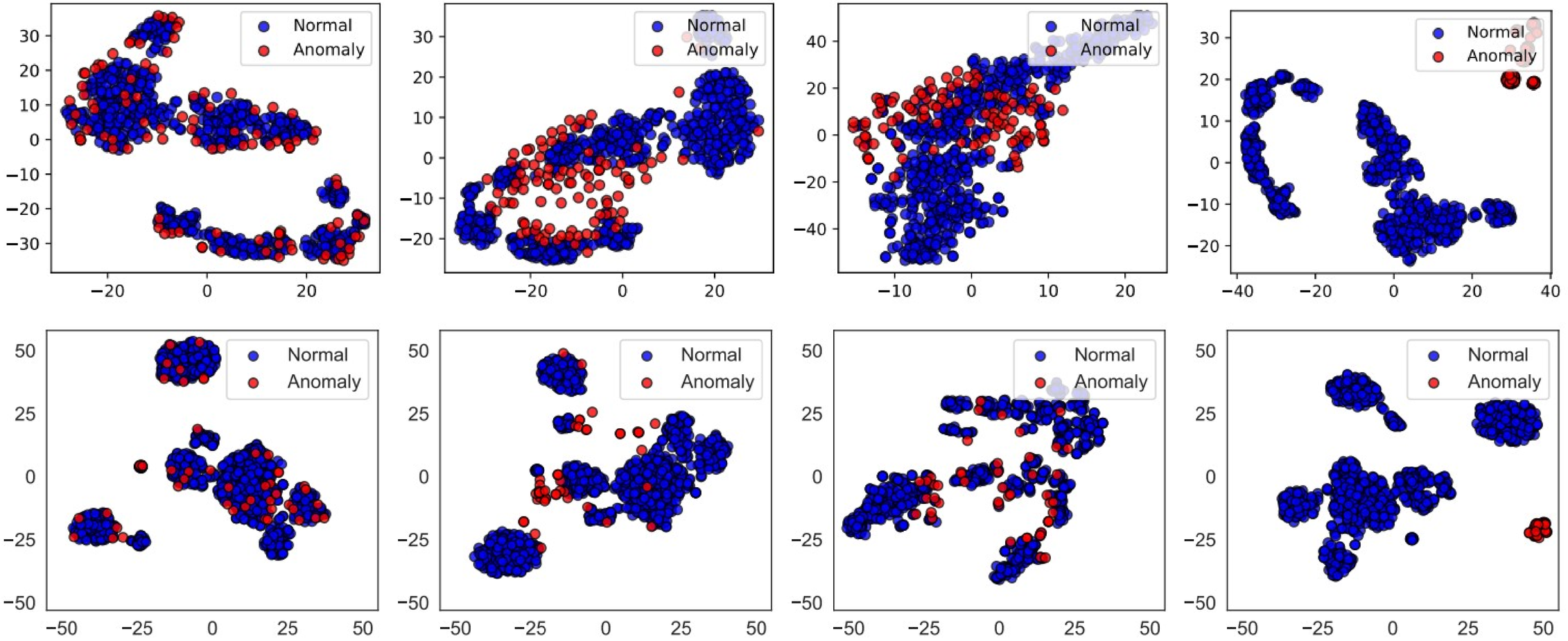








INTRODUCCIÓN

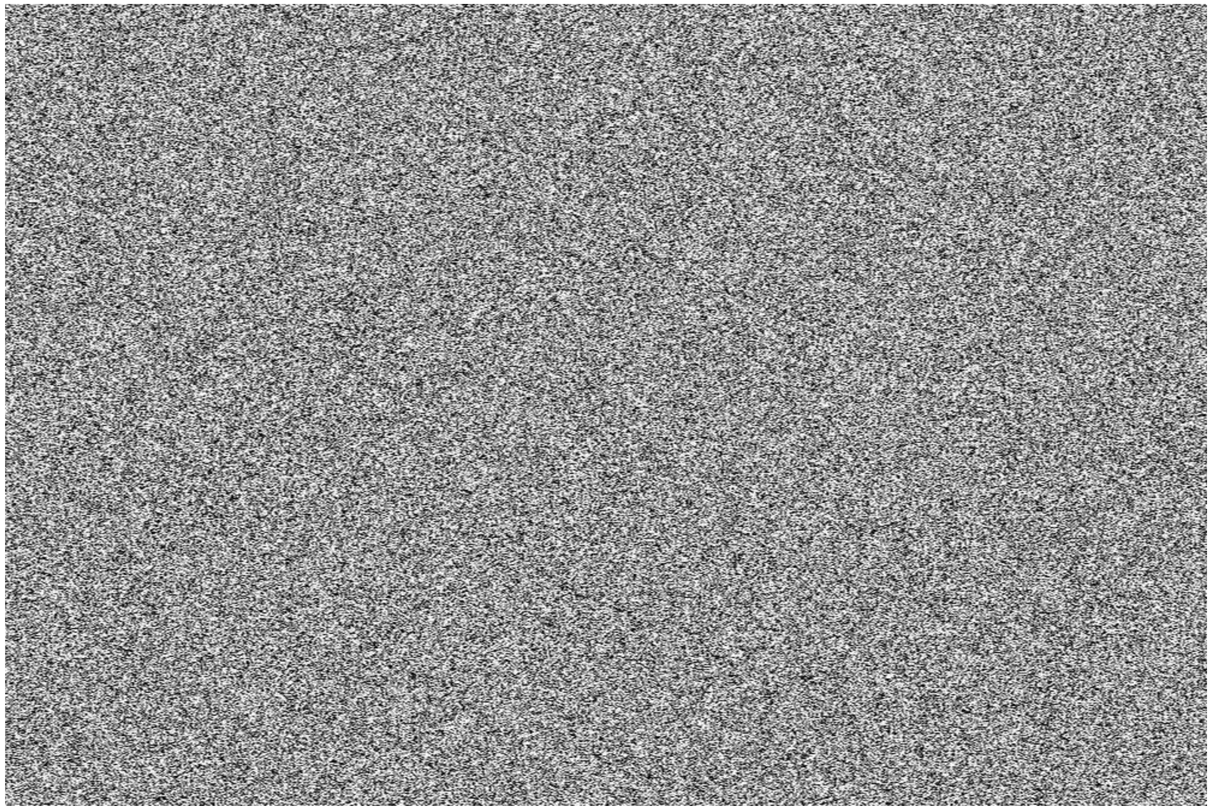


(a) Local (b) Global (c) Dependency (d) Clustered

Ruido

Definición

?



Ruido

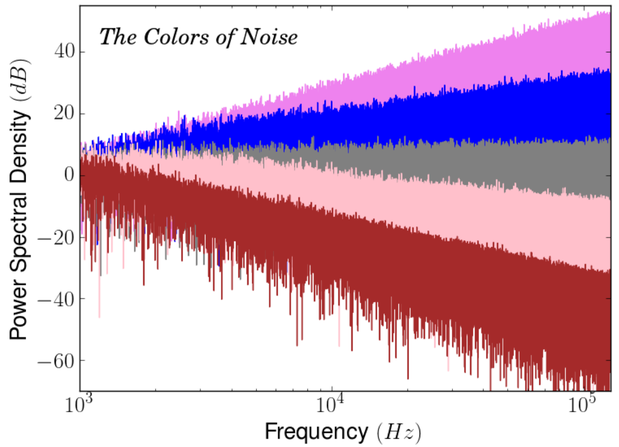
Definición

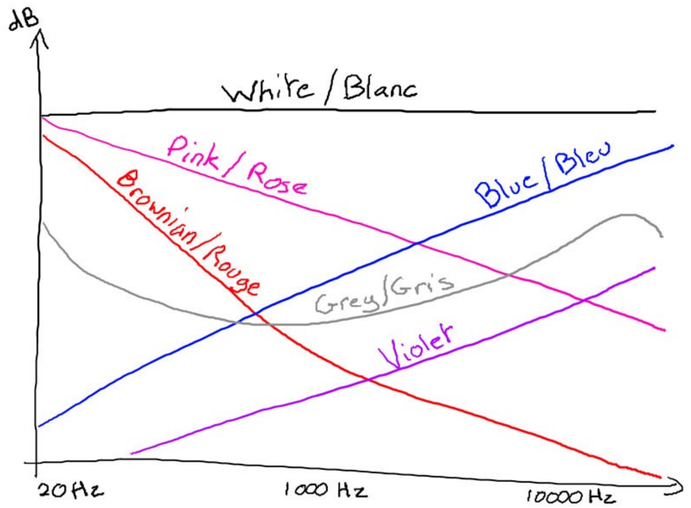
- White Noise (plano)

- Pink Noise (sonido)

Distribución estadística!

Ruido =? error





INTRODUCCIÓN

ENFOQUES

- No-supervisado
- Semi-supervisado
- Supervisado

INTRODUCCIÓN

SOLUCIONES

- Distancias
- Probabilidad
- Información
-

Regresión/Reconstrucción

- Kernels
- Redes

INTRODUCCIÓN

LIBRERIAS

- Scikitlearn
- pycaret
- Pyod

MAS: datos, libros, ejemplos...

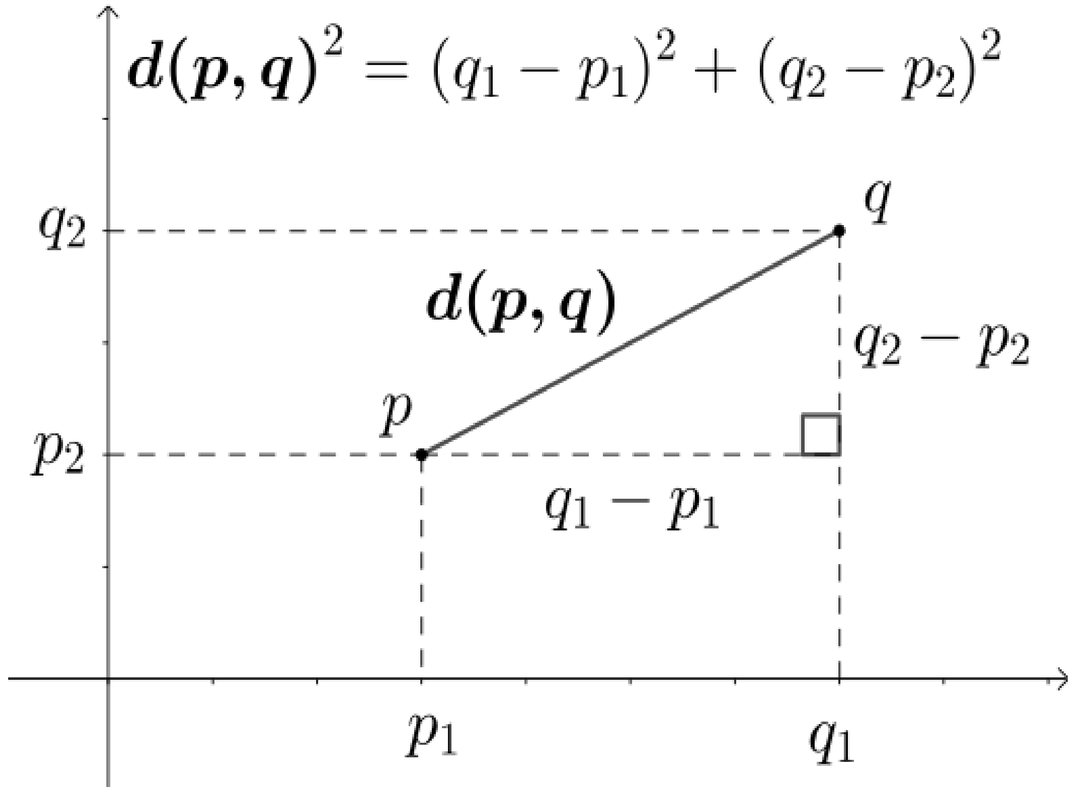
Distancias

Distancias

A mayor distancia, más anómalo

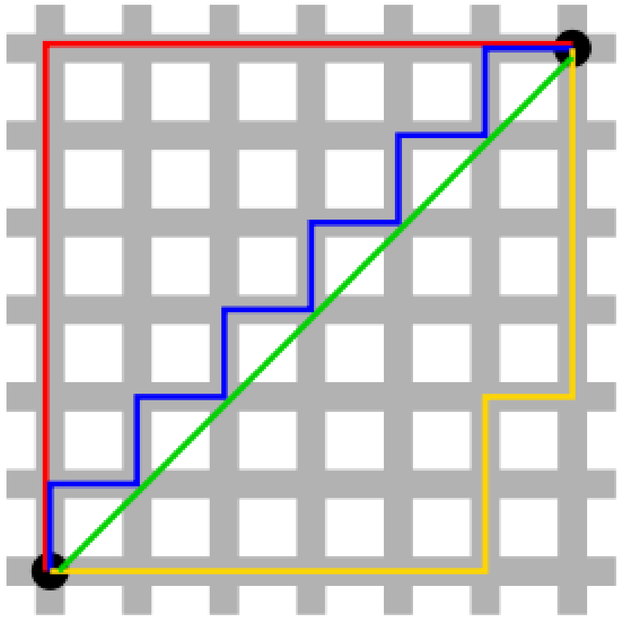
Distancias

Distancia Euclidea (L2, RMSE)



Distancias

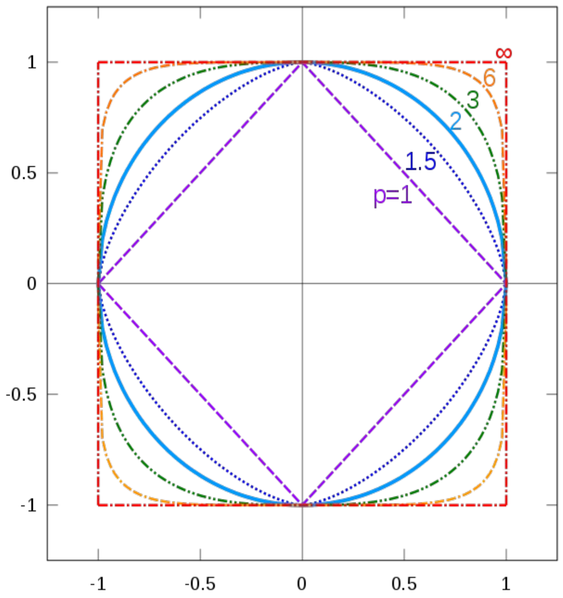
Distancia Manhattan (L1, MAE)



$$d_1(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_1 = \sum_{i=1}^{n} |p_i - q_i|$$

Distancias

Distancia p (Lp)

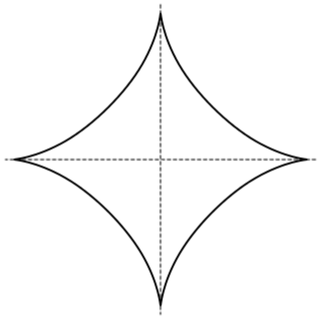


$$\|x\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{1/p}$$

Distancias

Distancia ∞ (L∞)

Distancia p<1

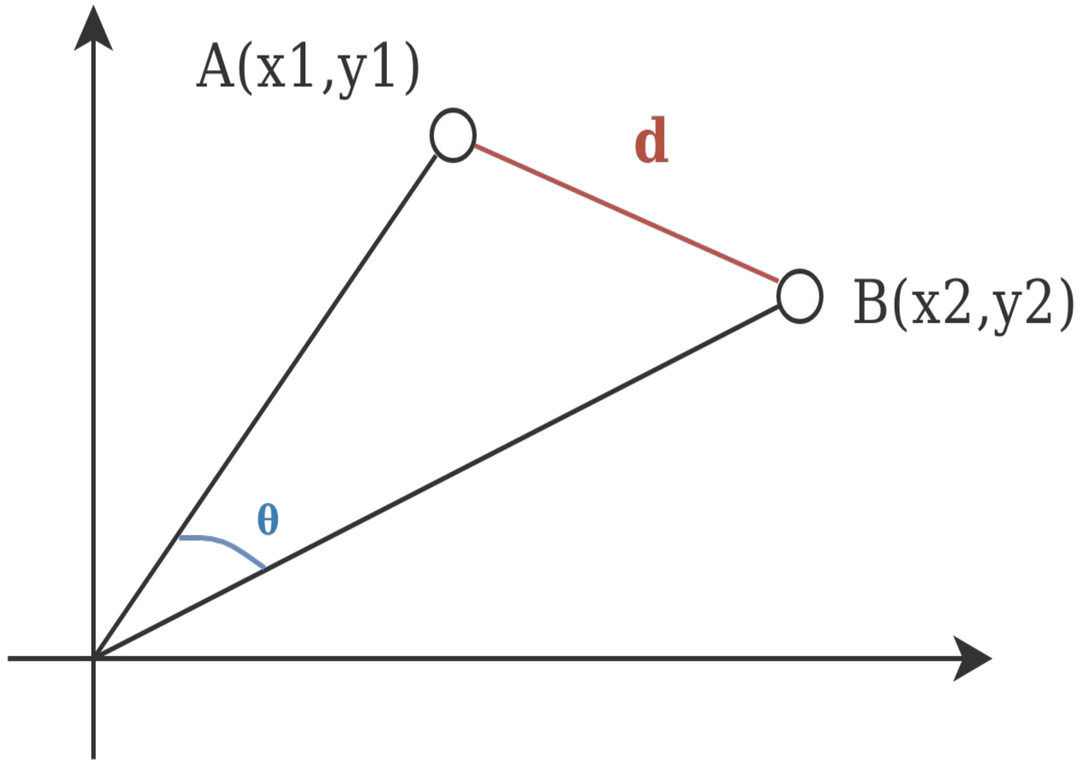


Distancia p=1 (L0)

Distancias

Distancia (o similitud) Coseno

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}$$

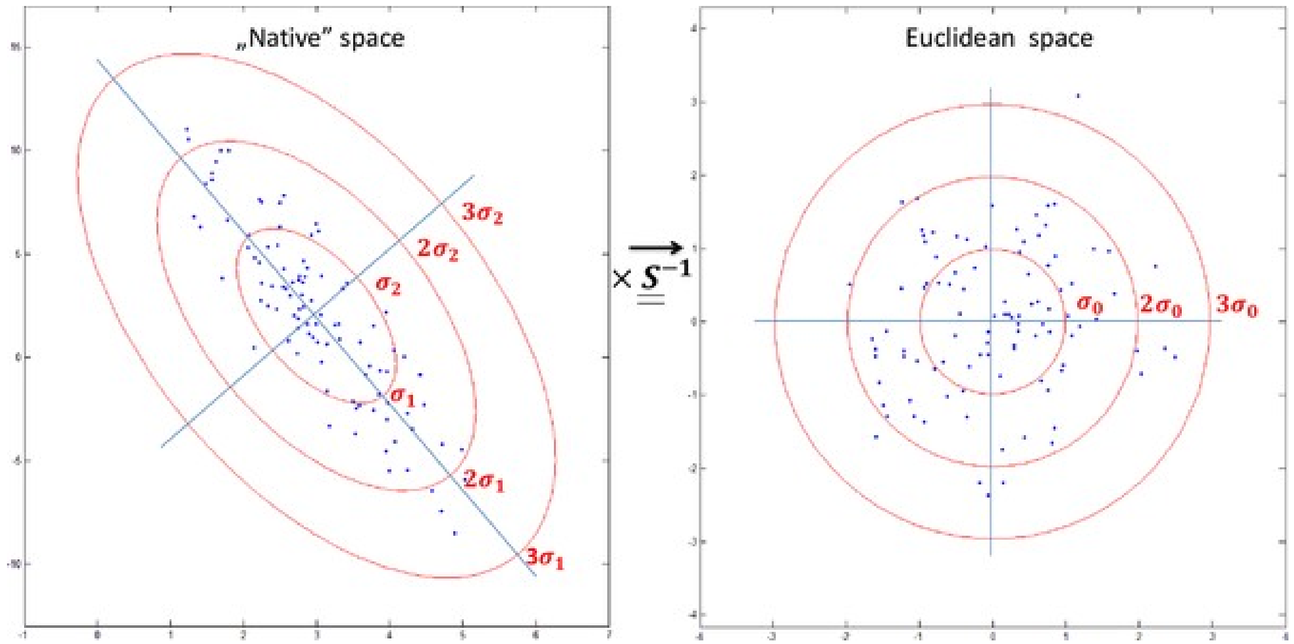


Distancias

Distancia Mahalanobis

$$d_M(\vec{x}, \vec{y}; Q) = \sqrt{(\vec{x} - \vec{y})^\mathsf{T} S^{-1} (\vec{x} - \vec{y})}$$

S Matriz de covarianza



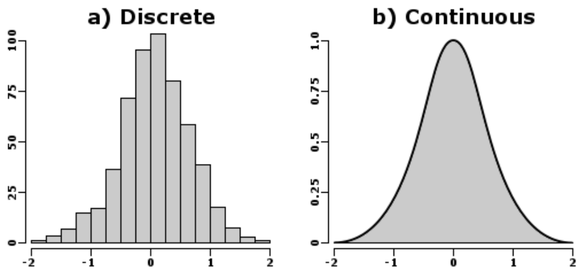
Probabilidad

Probabilidad

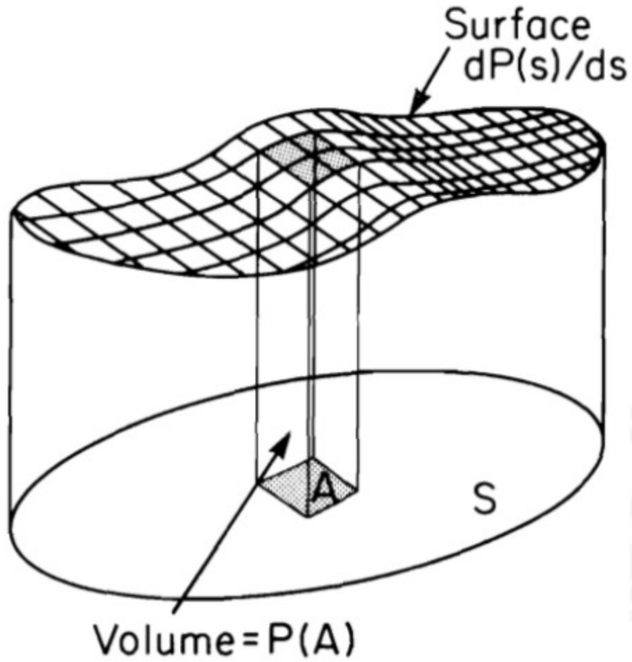
A menor probabilidad, más anómalo

Probabilidad

Discreta



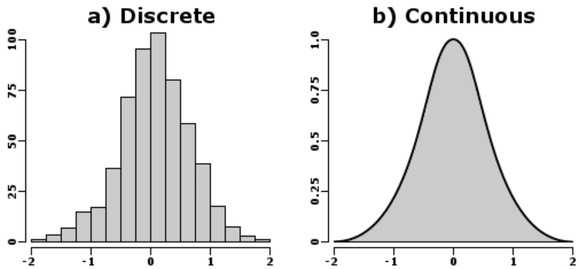
Continua



$$\Pr[a \leq X \leq b] = \int_a^b f_X(x)\,dx$$

Probabilidad

Discreta

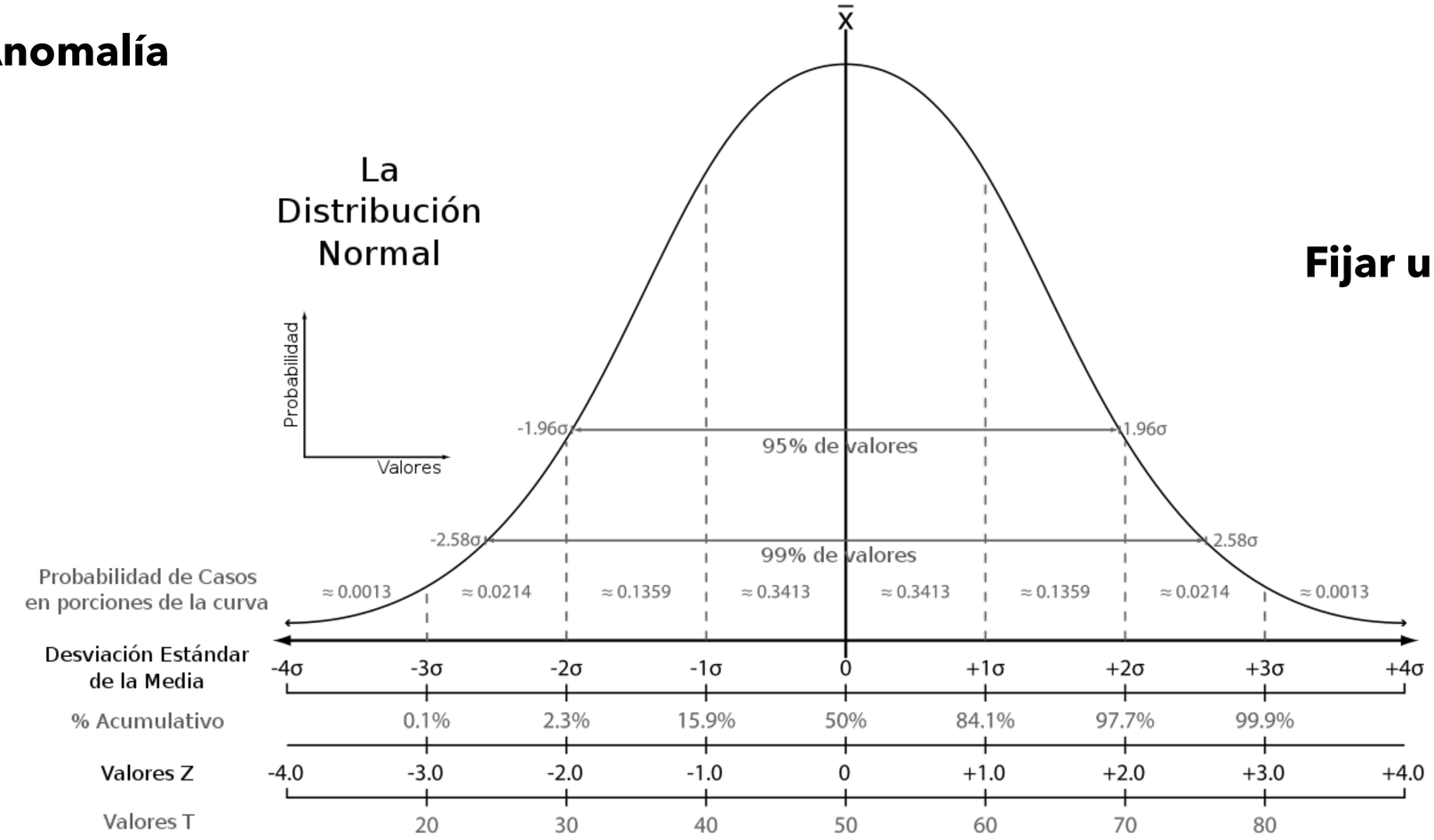


Continua

$$\Pr[a \leq X \leq b] = \int_a^b f_X(x)\,dx$$

	Discrete	Continuous
Probability Distribution	x_1 p_1 x_2 p_2 … … x_n p_n	pdf: f(x)
F(x)	$\sum_{i=1}^{n} p_i = 1$	$\int_{-\infty}^{\infty} f(x)dx = 1$
Mean μ	$\sum_{i=1}^{n} x_i p_i$	$\int_{-\infty}^{\infty} xf(x)dx$
Variance σ^2	$\sum_{i=1}^{n} (x_i-\mu)^2 p_i$	$\int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$

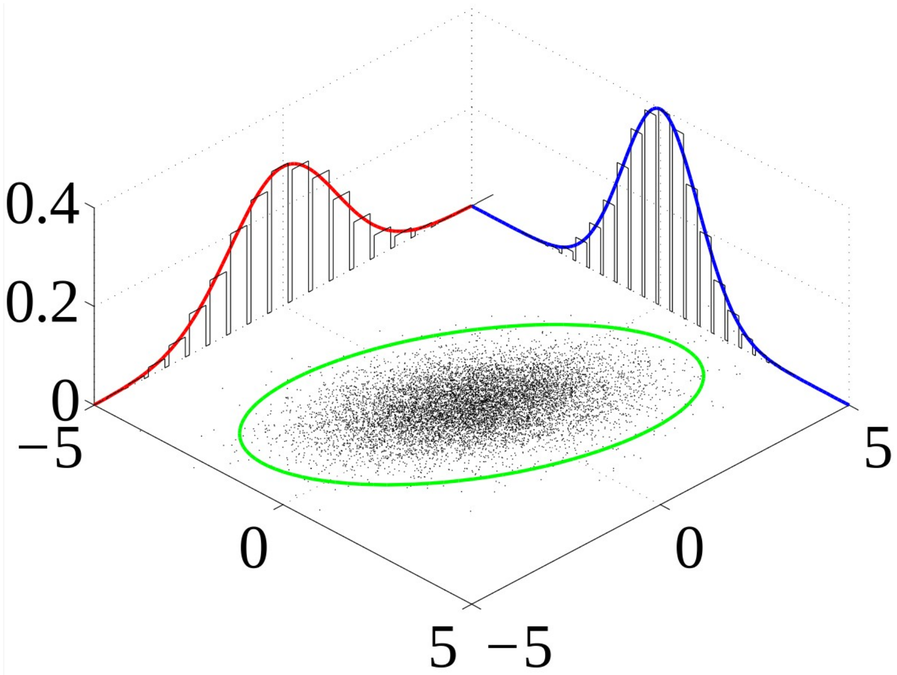
Anomalía

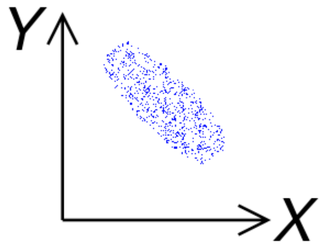


Fijar un número de sigmas

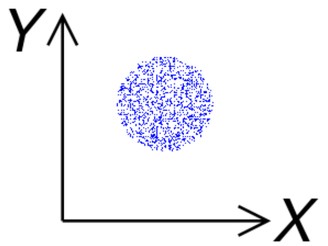
Probabilidad

Multidimensional:

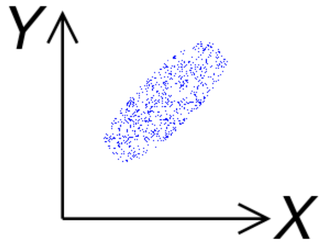




$\mathrm{cov}(X,Y) < 0$



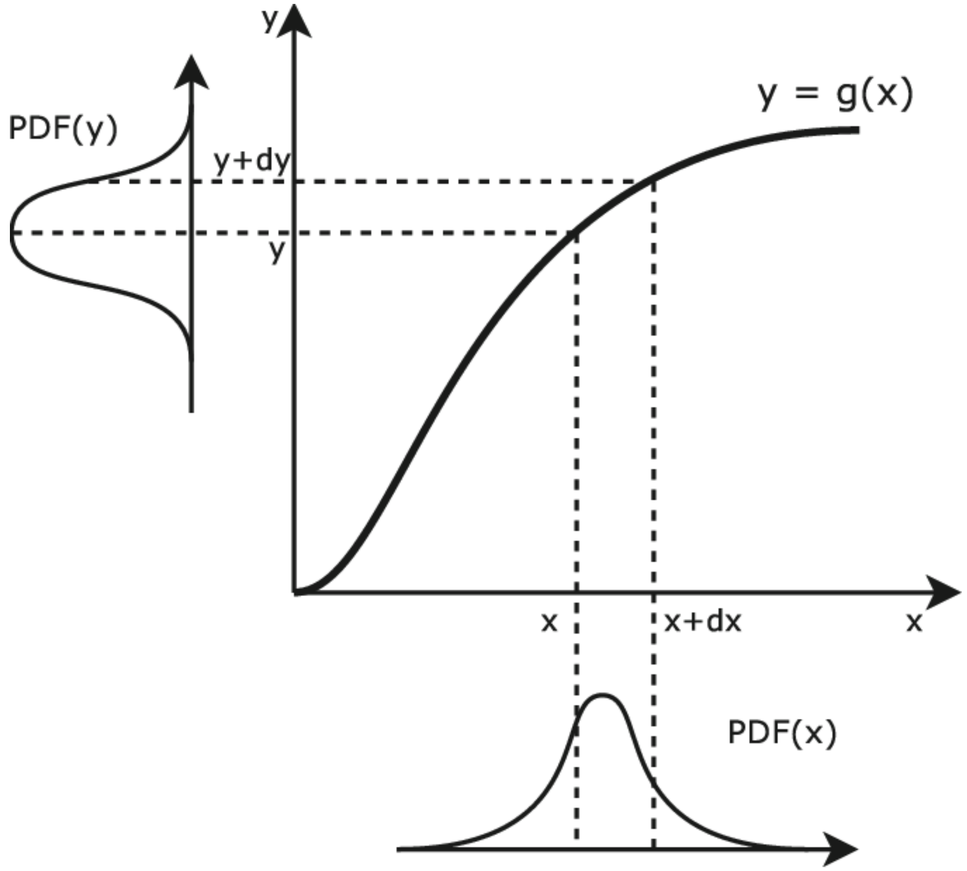
$\mathrm{cov}(X,Y) \approx 0$



$\mathrm{cov}(X,Y) > 0$

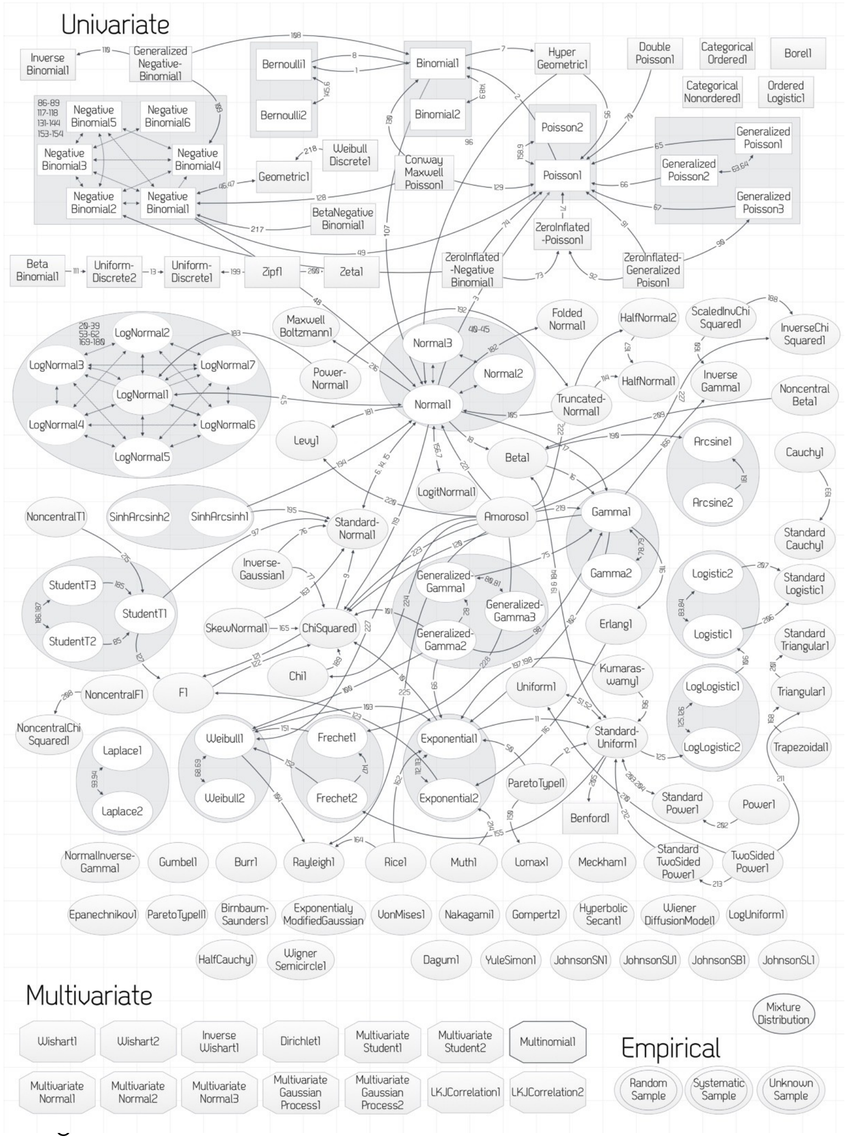
Probabilidad

Probabilidad bajo transformaciones:



Probabilidad

Modelos:



(o combinaciones)

https://en.wikipedia.org/wiki/List_of_convolutions_of_probability_distributions

Teoría de la información

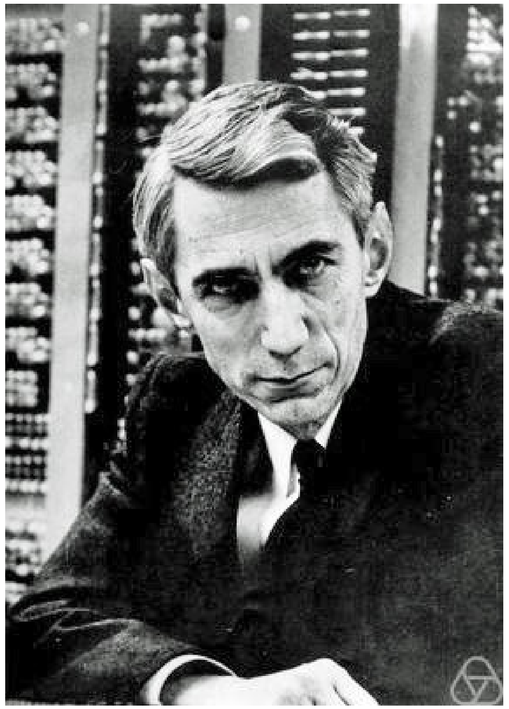
Teoría de la información

Información vs Anomalía

Teoría de la información

Información

Claude E. Shannon (1948) "A Mathematical Theory of Communication"



Información es proporcional a la inversa de la

$$I(x) \propto \frac{1}{p}(x)$$

Teoría de la información

Entropía

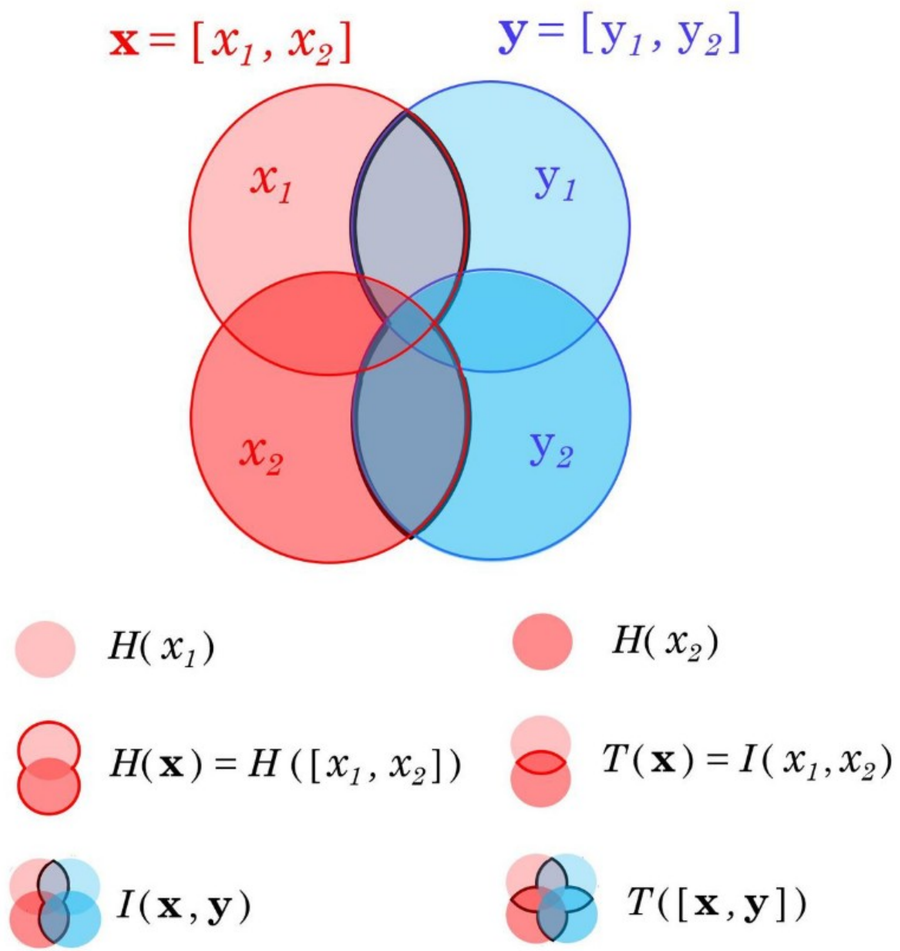
$$\mathrm{H}(X) = -\sum_{i=1}^{n} \mathrm{P}(x_i) \log \mathrm{P}(x_i)$$

Valor esperado de información de una fuente

Teoría de la información

Información mutua

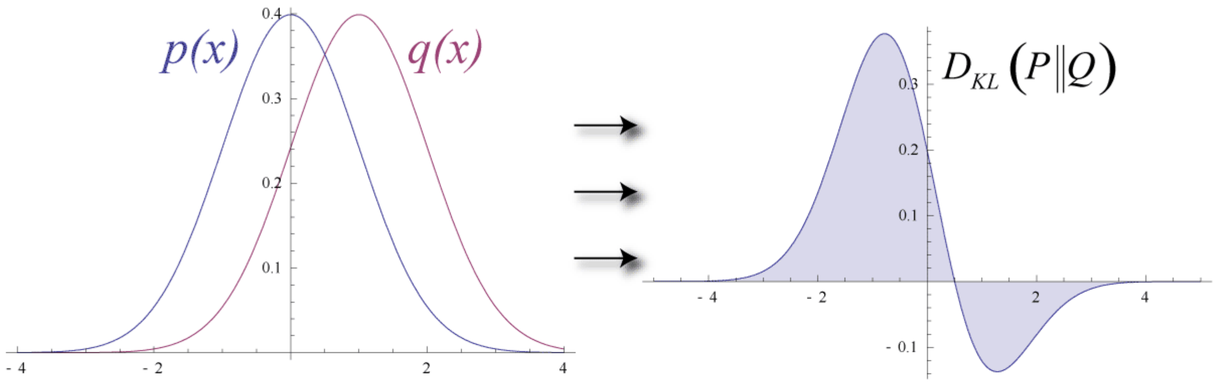
$$\mathrm{I}(X;Y) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p_{(X,Y)}(x,y) \log \left(\frac{p_{(X,Y)}(x,y)}{p_X(x)\, p_Y(y)} \right),$$



Teoría de la información

Divergencia de Kullback-Leibler (Relative entropy)

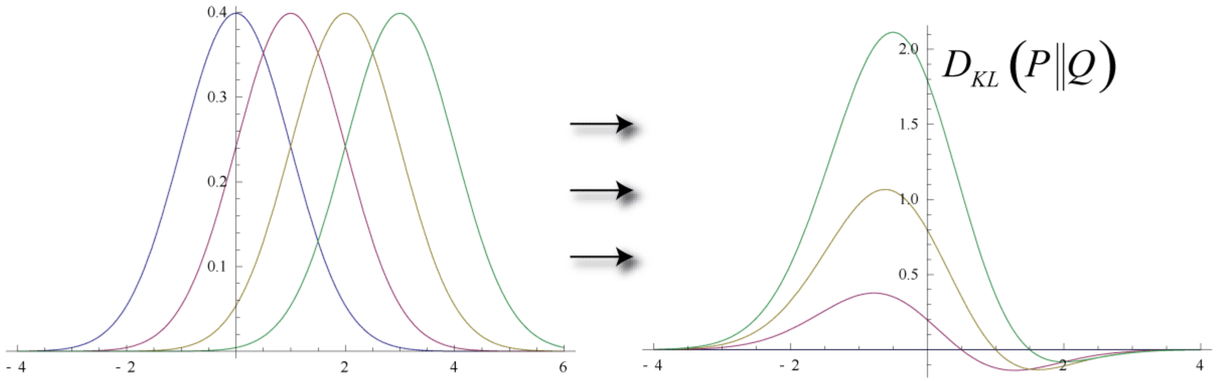
$$D_{\mathrm{KL}}(P\|Q) = \sum_i P(i) \ln \frac{P(i)}{Q(i)}.$$



Original Gaussian PDF's

KL Area to be Integrated

$$D_{\mathrm{KL}}(P \parallel Q) = -\sum_{x\in\mathcal{X}} p(x) \log q(x) + \sum_{x\in\mathcal{X}} p(x) \log p(x)$$
$$= \mathrm{H}(P,Q) - \mathrm{H}(P)$$



Teoría de la información

Miramos...

Information Theory Measures via Multidimensional Gaussianization

Valero Laparra, J. Emmanuel Johnson, Gustau Camps-Valls, Raul Santos-Rodríguez, Jesus Malo

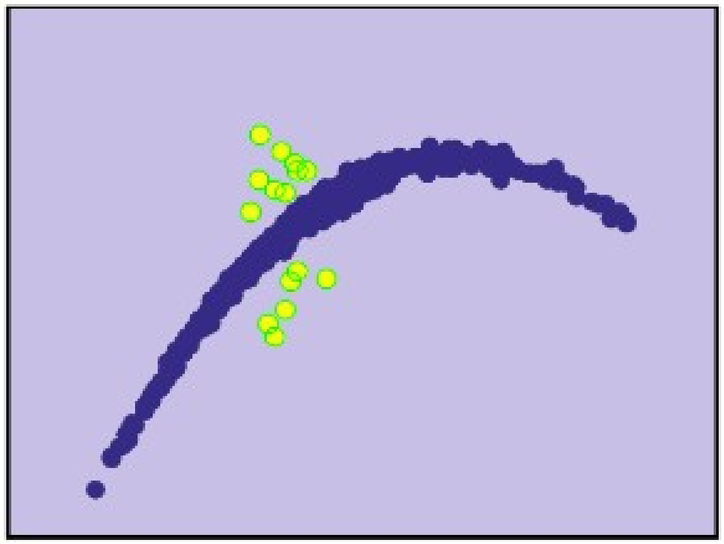
https://arxiv.org/abs/2010.03807

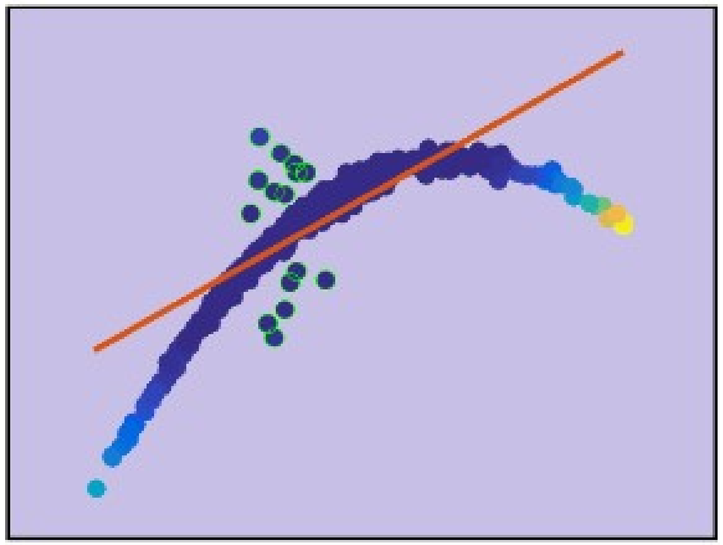
Regresión/ reconstrucción

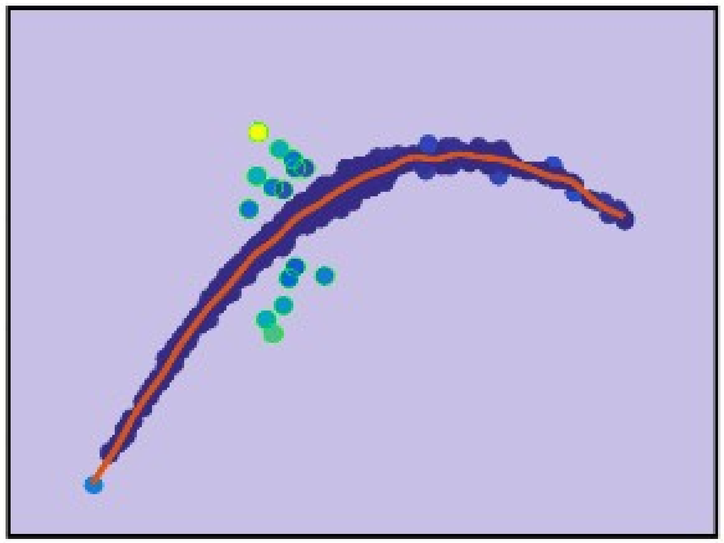
Regresión/reconstrucción

Mayor error de predicción, más anómalo.

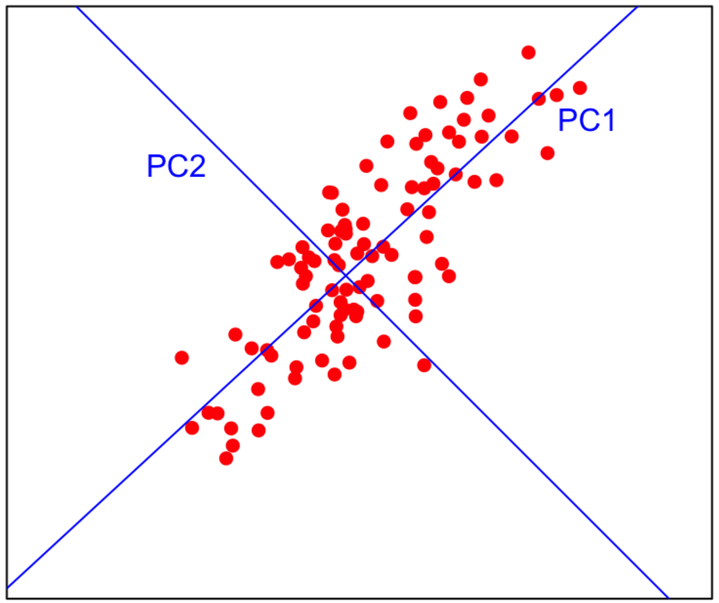
Regresión/reconstrucción



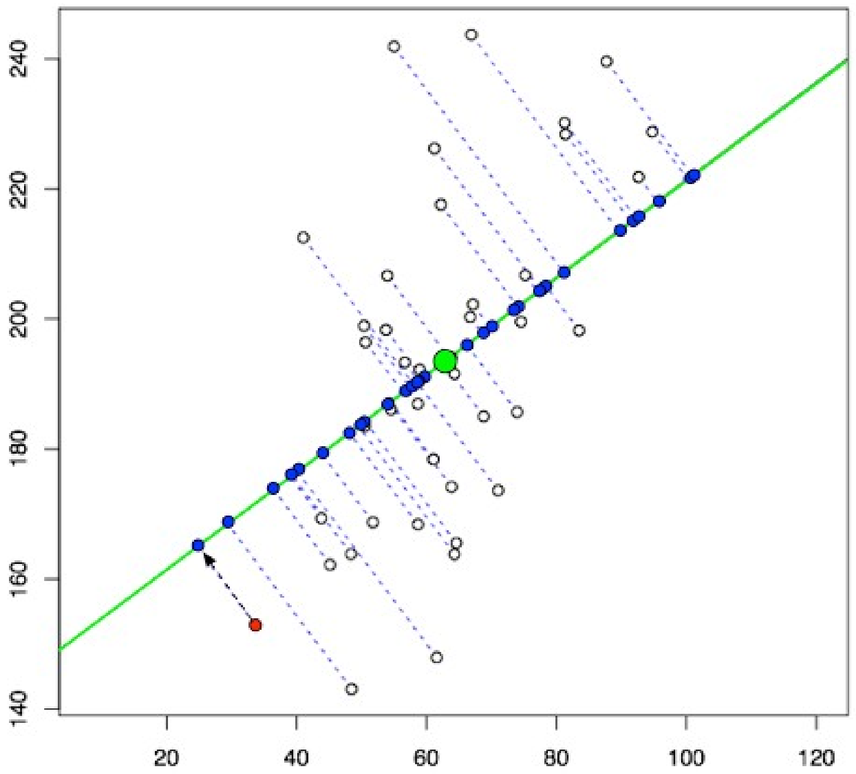




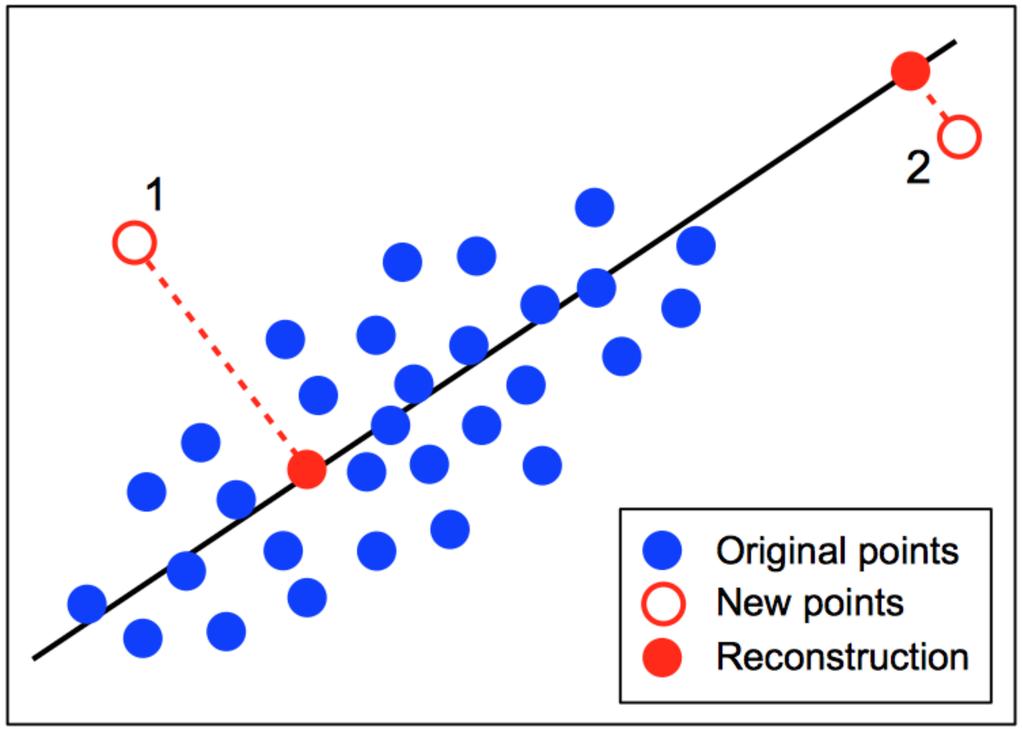
Regresión/reconstrucción



Regresión/reconstrucción



Regresión/reconstrucción



https://setosa.io/ev/principal-component-analysis/

Algoritmos clásicos

INDICE

- K-means, VQ y GMM

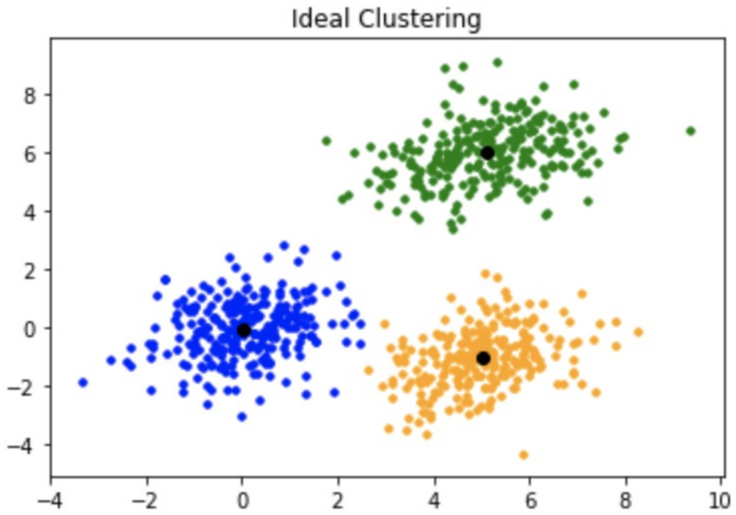
- DBSCAN (1996)

- LOF (2000)

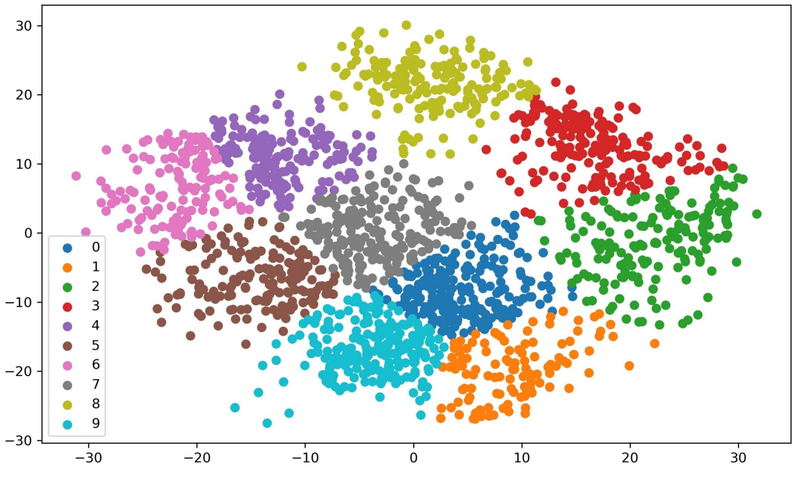
- Isolation Forest (2008)

K-means

Clustering

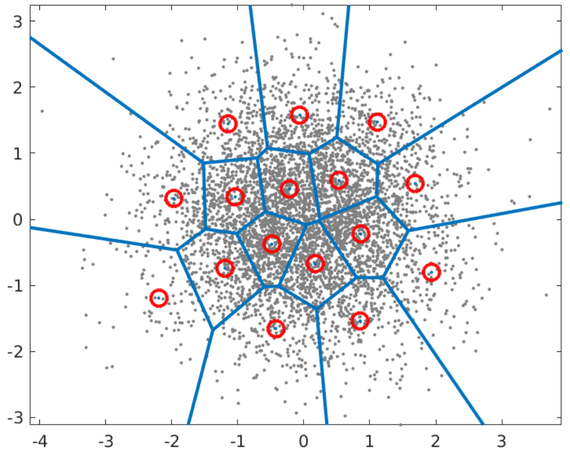






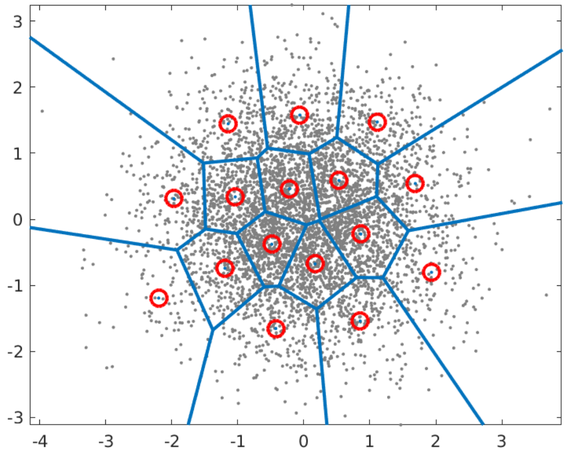
K-means

Vector Quantization



K-means

Vector Quantization



En la práctica:

- Distintas aproximaciones:
  i.e. Lloyds (1-asigna y 2-calcula la media)
- Depende de la inicialización
  i.e. random

https://clustering-visualizer.web.app/kmeans

https://en.wikipedia.org/wiki/K-means_clustering

DBSCAN

Density-based spatial clustering of applications with noise
https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf

- Punto núcleo (A)

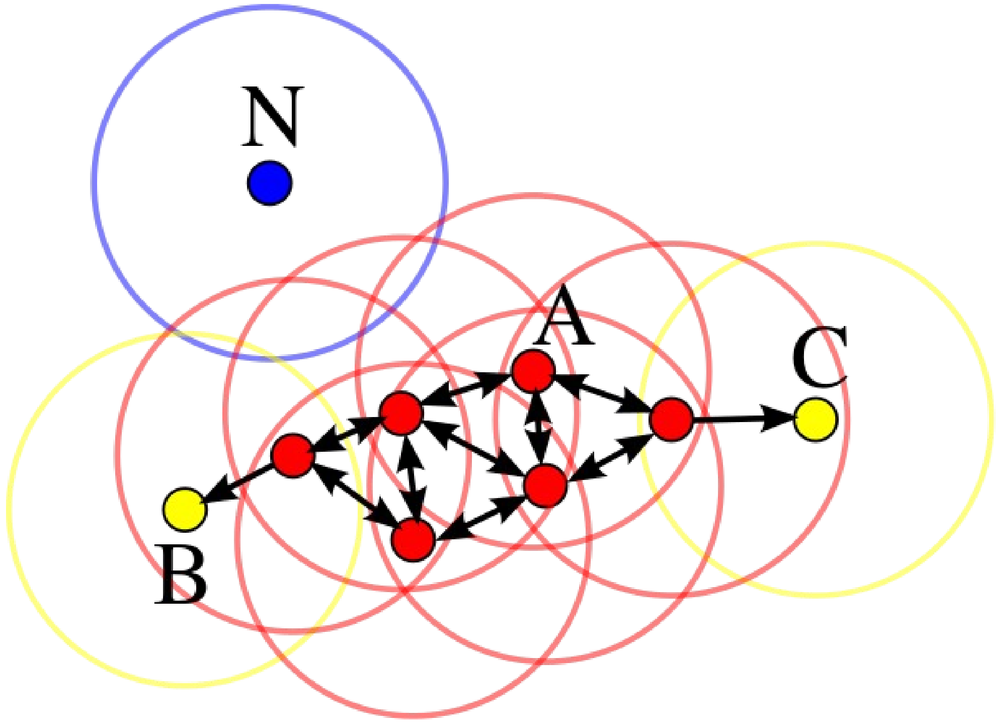
Si **n** puntos estan a **ε**

- Puntos (densamente-)alcanzables (B y C)

Si no son punto núcleo,
pero están a una distancia menor de **ε** de un punto núcleo

- Ruido (N)

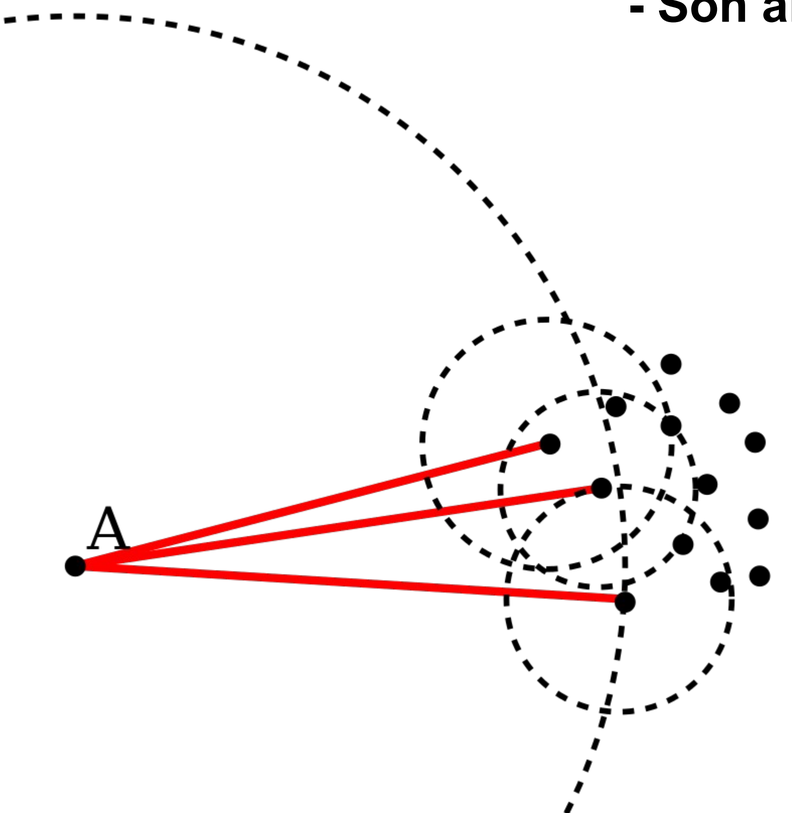
Si no es alcanzable



LOF

local outlier factor (LOF)
https://www.dbs.ifi.lmu.de/Publikationen/Papers/LOF.pdf

- Son anomalias los puntos que tienen una densidad menor que sus vecinos.

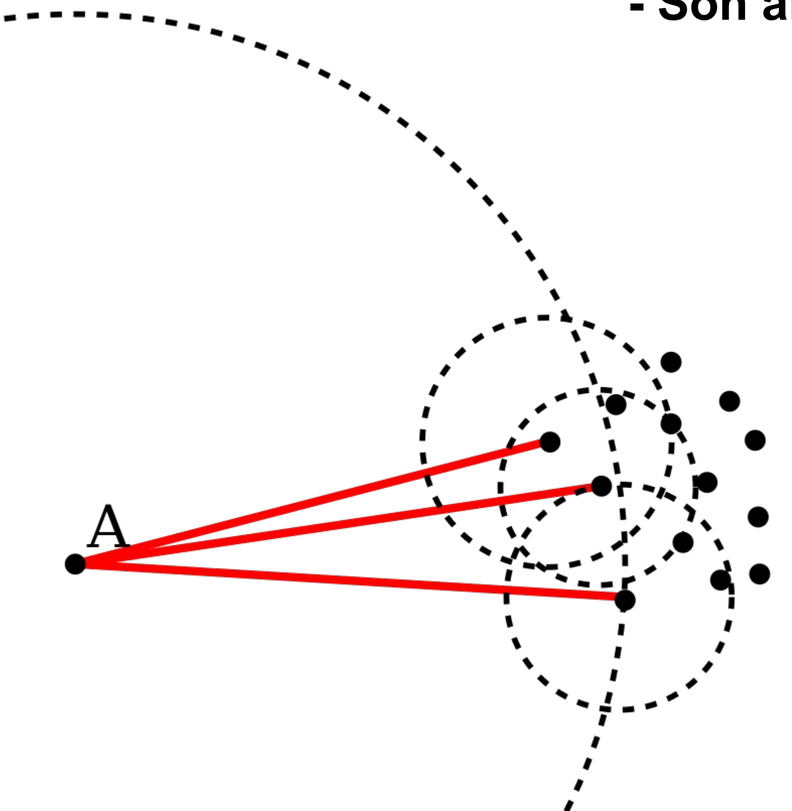


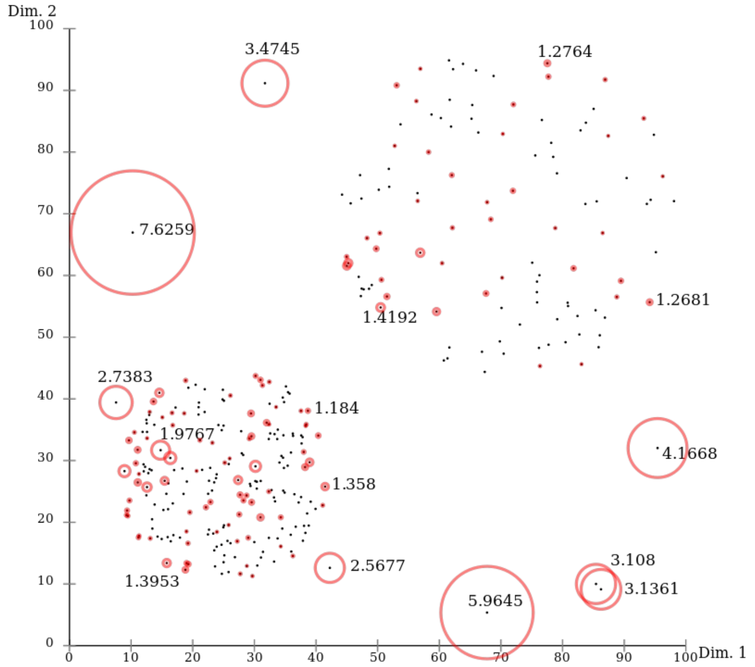
https://en.wikipedia.org/wiki/Local_outlier_factor

LOF

local outlier factor (LOF)
https://www.dbs.ifi.lmu.de/Publikationen/Papers/LOF.pdf

- Son anomalias los puntos que tienen una densidad menor que sus vecinos.

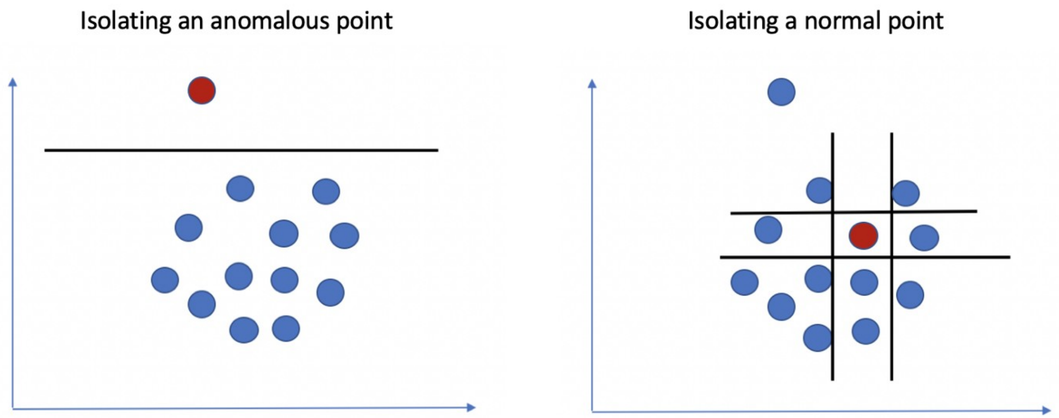




Isolation Forest

Isolation Forest
https://ieeexplore.ieee.org/document/4781136/

- Detecta anomalias usando árboles de decisión.
Un punto es anómalo si se le puede aislar con pocos cortes (decisiones).

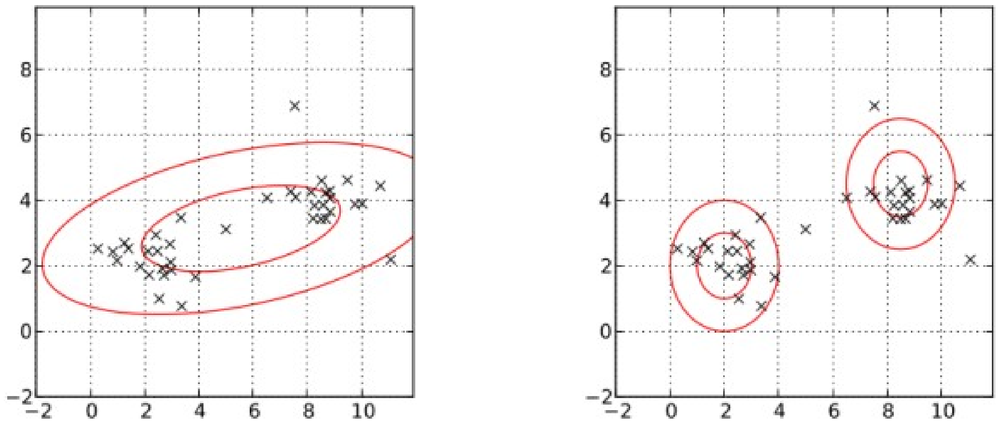


https://en.wikipedia.org/wiki/Isolation_forest

https://www.cienciadedatos.net/documentos/py22-deteccion-anomalias-isolation-forest-python.html

GMM

GMM Gaussian Mixure Model

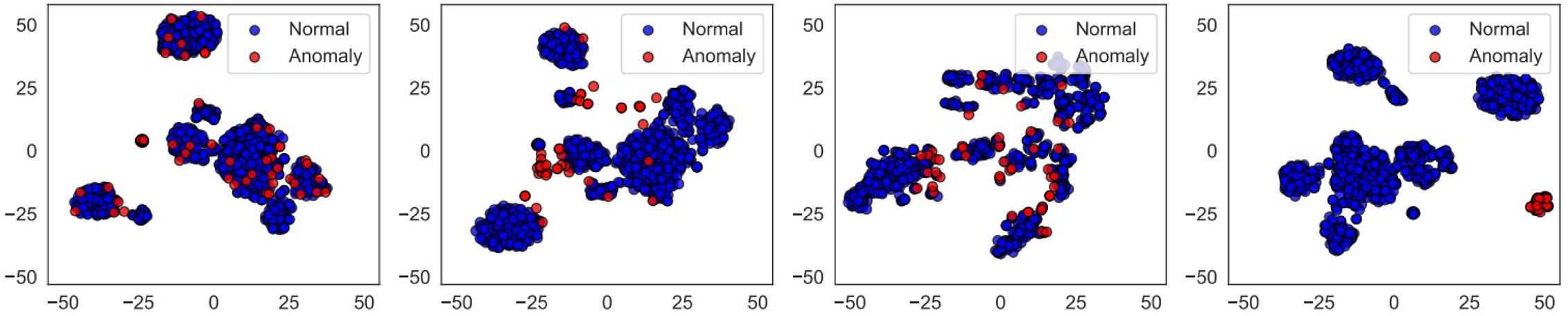


En la práctica:
- Expectation maximization (EM)
- Maximum likelihood

https://www.cienciadedatos.net/documentos/py23-deteccion-anomalias-gmm-python.html

Comparativa

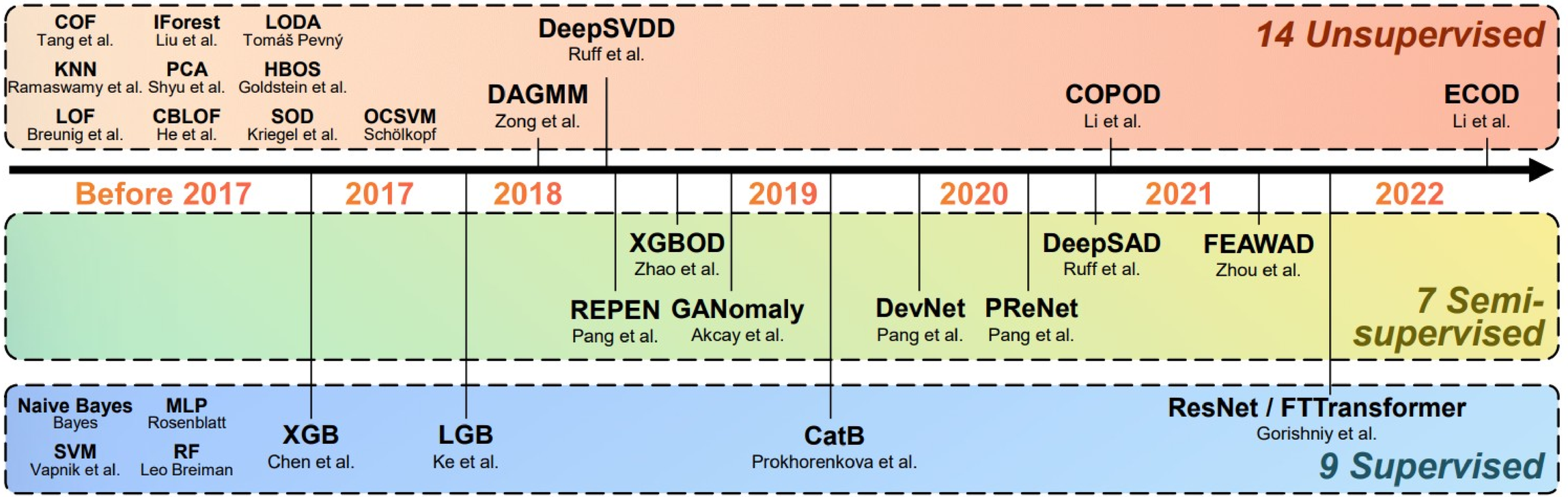
COMPARATIVA



(a) Local (b) Global (c) Dependency (d) Clustered

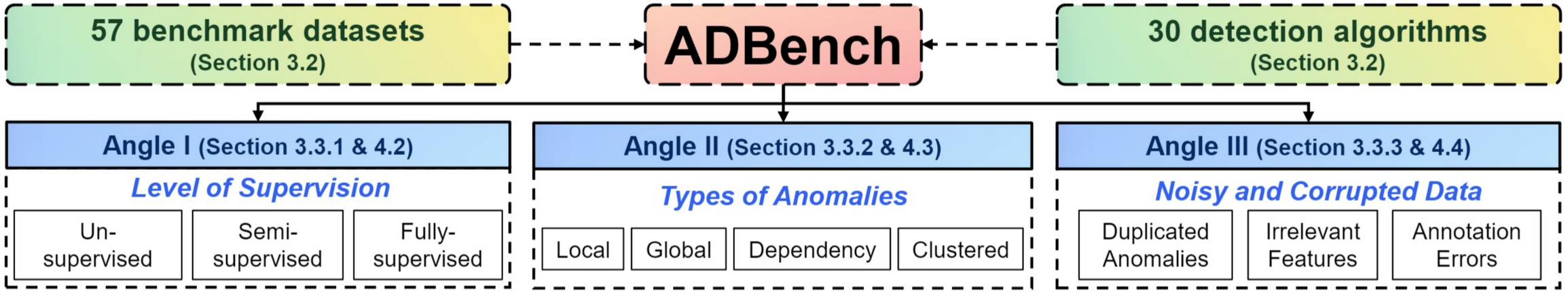
ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf

COMPARATIVA



ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://arxiv.org/abs/2206.09426

COMPARATIVA



DATOS: https://github.com/Minqi824/ADBench

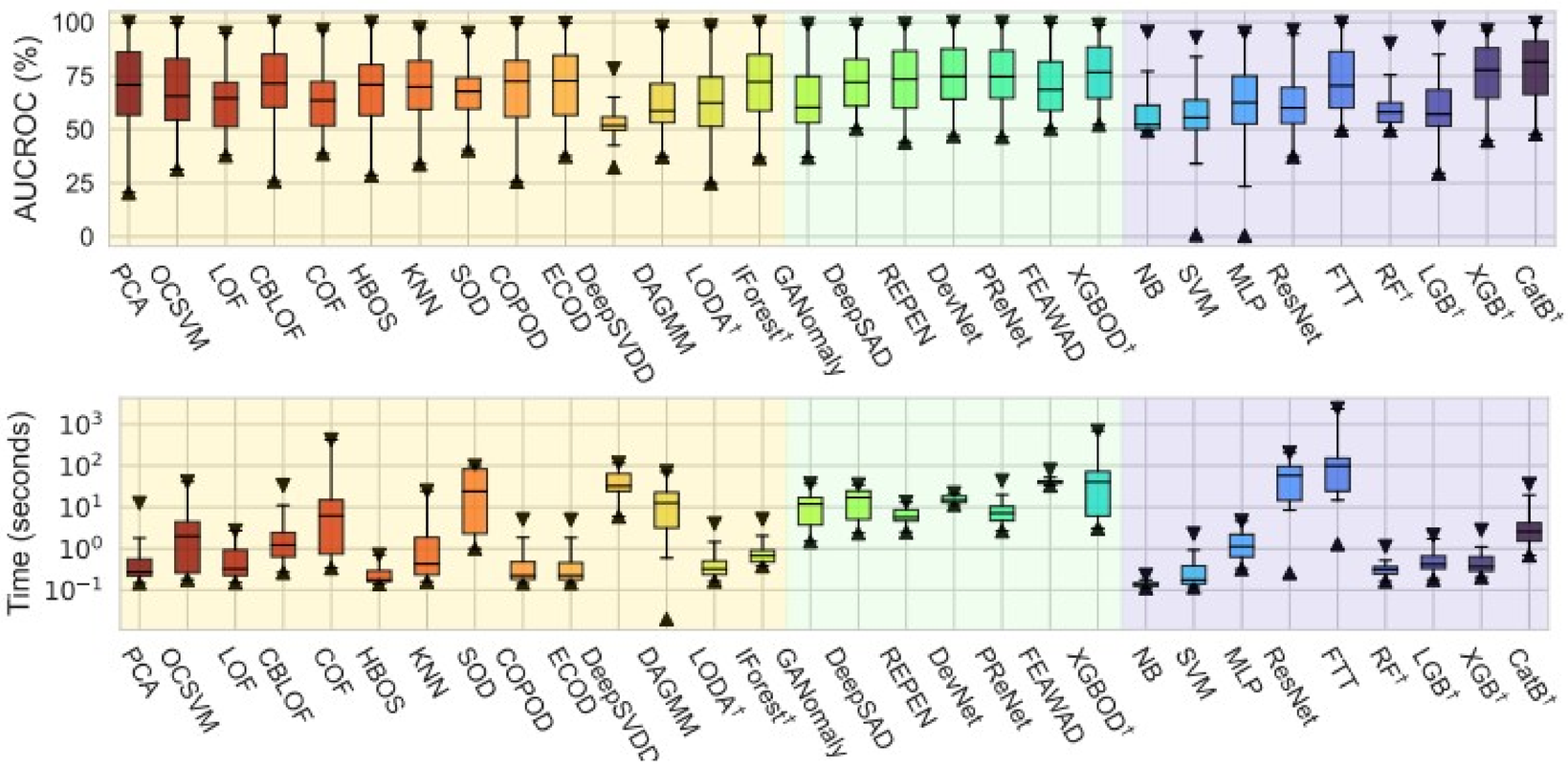
ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf

COMPARATIVA

RESULTADOS

- 57 datasets
- 98,436 experimentos

- No-supervisados
- Semi-supervisados
- Supervisados

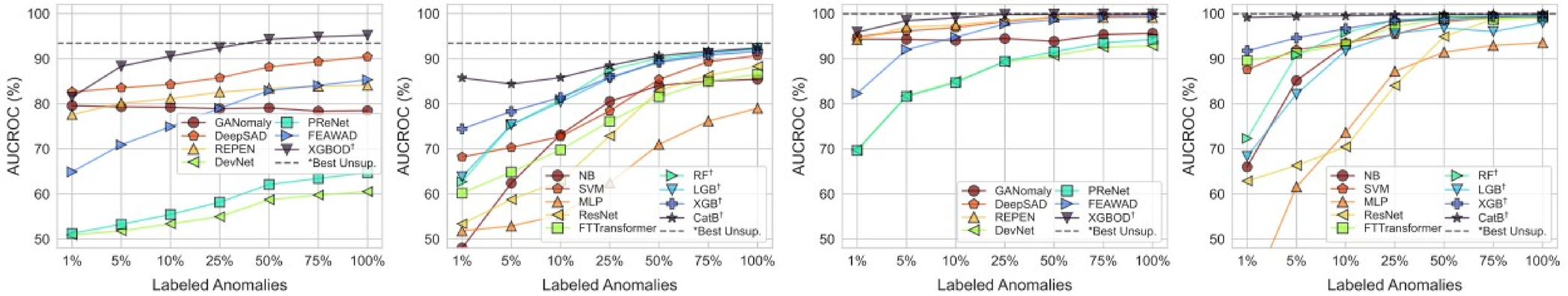


DATOS: https://github.com/Minqi824/ADBench

ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf

COMPARATIVA

RESULTADOS (mejor no supervisado en discontinua)



(a) Local anomalies

(b) Global anomalies

(c) Dependency anomalies

(d) Clustered anomalies

COMPARATIVA

CONCLUSIONES (no-supervisados)

- “None of the unsupervised methods is statistically better than the others.”

- “Some DL-based unsupervised methods like DeepSVDD and DAGMM are surprisingly worse than shallow methods.”

- “DL-based unsupervised algorithms are harder to train”

- “Future algorithms should be evaluated on large testbeds like ADBench for statistical tests”

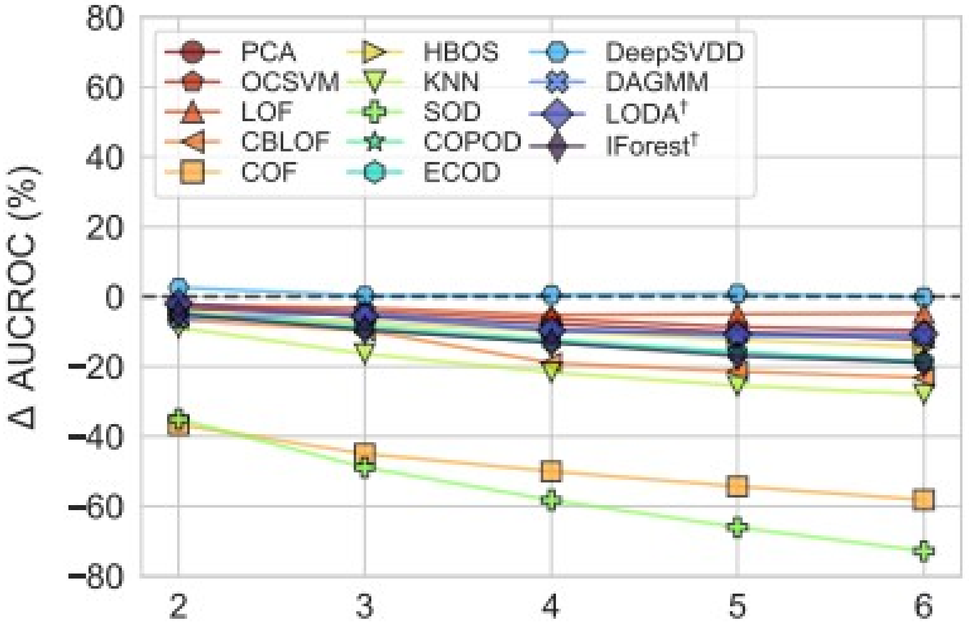
- “The **no-free-lunch theorem**, suggests there is no universal winner for all tasks”

ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf

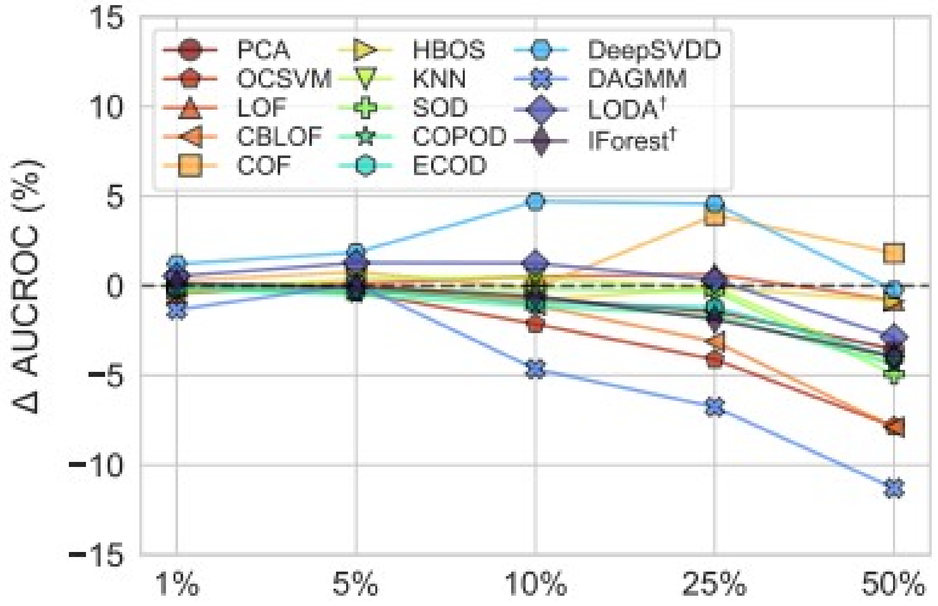
COMPARATIVA

RESULTADOS con datos corruptos

Datos duplicados



Características irrelevantes



ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf

COMPARATIVA

CONCLUSIONES (no-supervisados)

- “All unsupervised methods are severely impacted by duplicated anomalies.”

- “Unsupervised methods often assume the underlying data is imbalanced with only a smaller percentage of anomalies.”

- “Incorporate unsupervised feature selection”

ADBench: Anomaly Detection Benchmark. Songqiao Han, et al. Neurips 2022
https://www.andrew.cmu.edu/user/yuezhao2/papers/22-neurips-adbench.pdf